

mbed Simulator

ARM mbed OS is a free, open source embedded operating system designed for various ARM Cortex-M based targets. It comes with a rich set of drivers for various sensors, I/O devices, connectivity with a powerful networking stack, RTOS APIs and security. Recently, a Web based simulator has been introduced. It can run various examples with sensor interfacing and networking through mbed APIs.

URL: <https://labs.mbed.com/simulator>

Interactive notebooks and IDEs

Jupyter Notebook

This is an open source Web application, formally known as IPython. It allows users to create and share documents

with live code, equations, graph plotting and documentation. Though Jupyter is designed primarily for Python, it supports various kernels for C, C++, Java, JavaScript and other languages. Xeus-cling is a Jupyter kernel for C++, based on the Cling interpreter. For engineering education, it provides an interactive way to learn complex algorithms and fundamentals in different programming languages through Notebooks.

URLs: <https://github.com/jupyter/jupyter/wiki/Jupyter-kernels>; <https://xeus-cling.readthedocs.io/en/latest/>

PythonTutor

This helps in overcoming a fundamental barrier to learn programming by running code line by line, by code tracing, and the visualisation of the stack and heap. The Web service allows visualisation of code snippets that can be embedded into other services through Iframe. It supports other programming languages apart from Python, like Java, C, C++, JavaScript and Ruby. More than five million people have used the service to visualise over 100 million pieces of code, often as a supplement to textbooks, lectures and online tutorials.

URL: <http://pythontutor.com/> 

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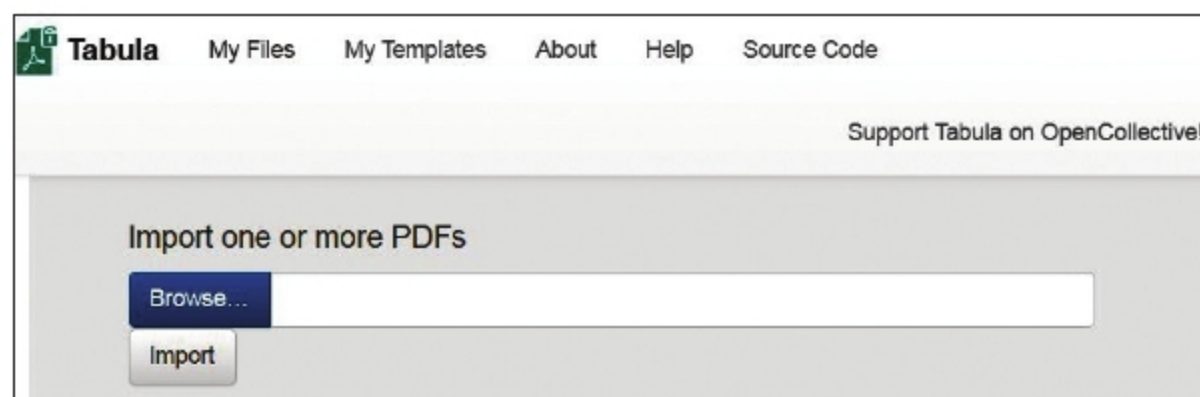


Figure 4: Importing the PDF files on the Tabula dashboard



Figure 5: Preview and export the tables from PDF files

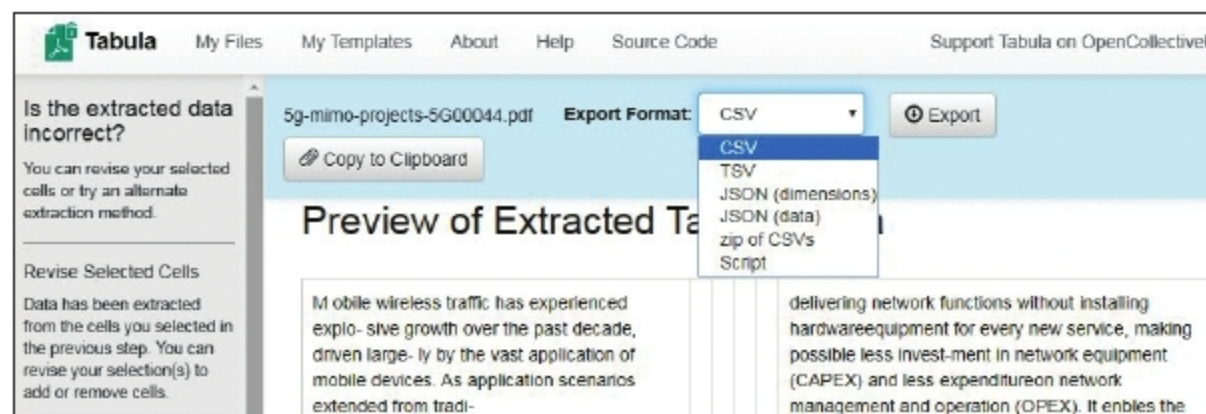


Figure 6: Transformation of fetched tables in multiple formats

Tabula can be used by research scholars and data scientists to fetch selected tables from research papers and government reports for use in multiple applications. Research papers and analytics reports are generally prepared as PDF files and are available on various portals for users. With the help of Tabula, data sets from the PDF files can be extracted and converted to the CSV or Excel formats so that the data mining or machine learning algorithms can be directly implemented. Also, once the data is extracted to the CSV format, it can be plotted for multiple applications. 

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The author is the MD of Magma Research and Consultancy Services, Ambala. He is associated with various universities, institutes and autonomous organisations, where he gives lectures, conducts technical workshops and offers consultancy on the latest technologies and tools.